EDA Report

AI Summary

The provided dataset, consisting of 891 rows and 12 columns, shows an 8.1% total missing value rate, predominantly in the 'Age', 'Cabin', and 'Embarked' columns. Notable distributions include a skewed distribution for 'Fare' with an excess of high values, and a bimodal distribution for 'Survived' with a mean of 38.38%, indicating a substantial proportion of survivors but still an imbalance towards non-survivors. Significant correlations were found between 'Pclass' and 'Fare', 'Pclass' and 'Age', and 'SibSp' and 'Parch', providing insight into the relationships between these variables. Additionally, the presence of outliers in 'Age', 'SibSp', 'Parch', and 'Fare' suggests data quality issues that require investigation and potential data imputation or transformation to ensure accurate analysis.

1. Dataset Overview

Metric
Rows
Columns
Total Cells
Duplicate Rows
Memory Usage

Column Details

Column	Type	Nulls	Null %	Unique
PassengerId	int64	0	0%	891
Survived	int64	0	0%	2
Pclass	int64	0	0%	3
Name	str	0	0%	891

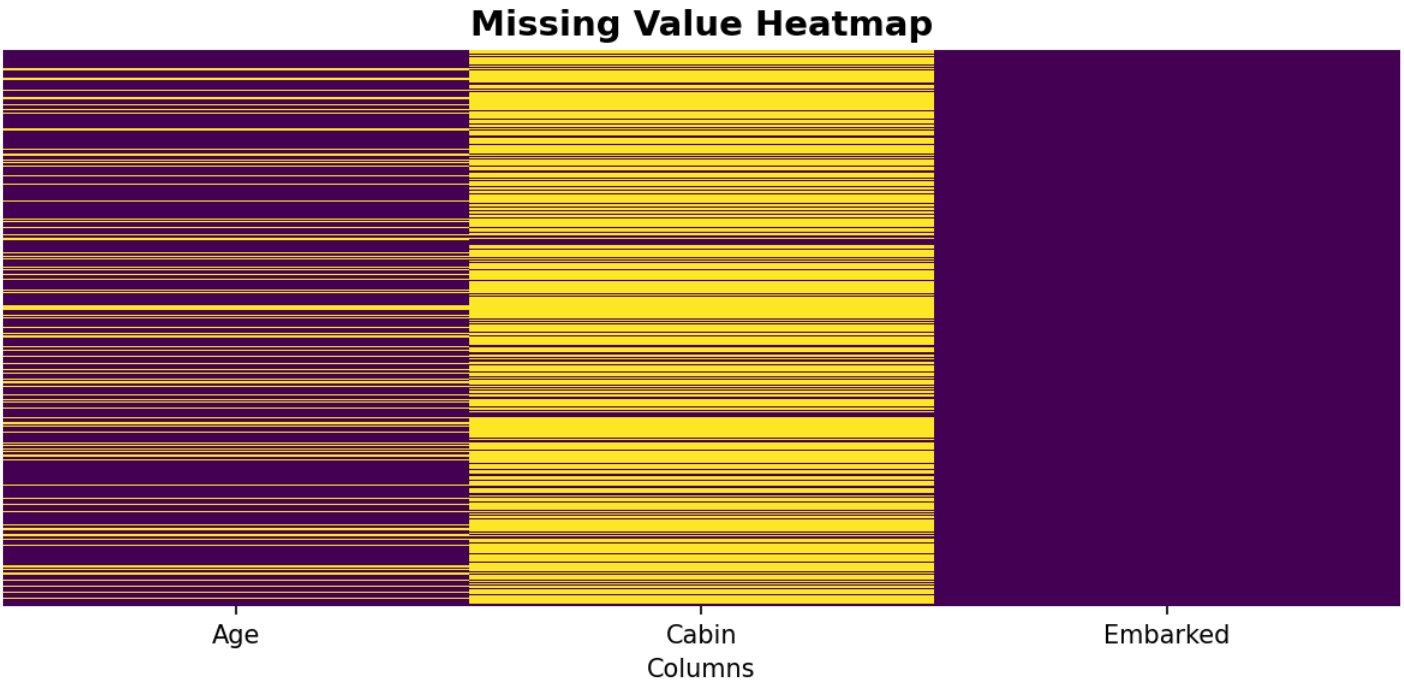
Sex	str	0	0%	2
Age	float64	177	19.87%	88
SibSp	int64	0	0%	7
Parch	int64	0	0%	7
Ticket	str	0	0%	681
Fare	float64	0	0%	248
Cabin	str	687	77.1%	147
Embarked	str	2	0.22%	3

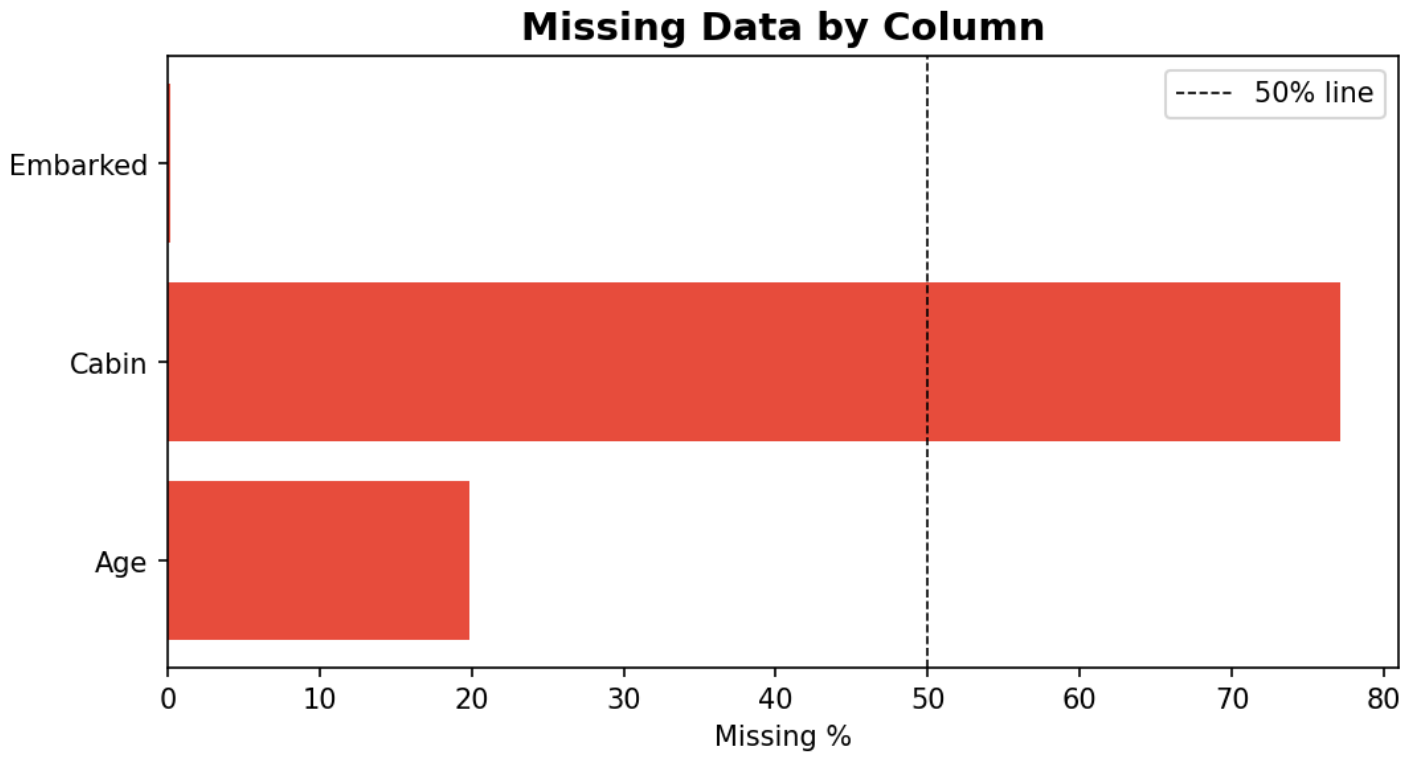
2. Missing Data Analysis

Metric	Value
Total Missing Cells	866
Overall Missing %	8.1%

Columns with Missing Values

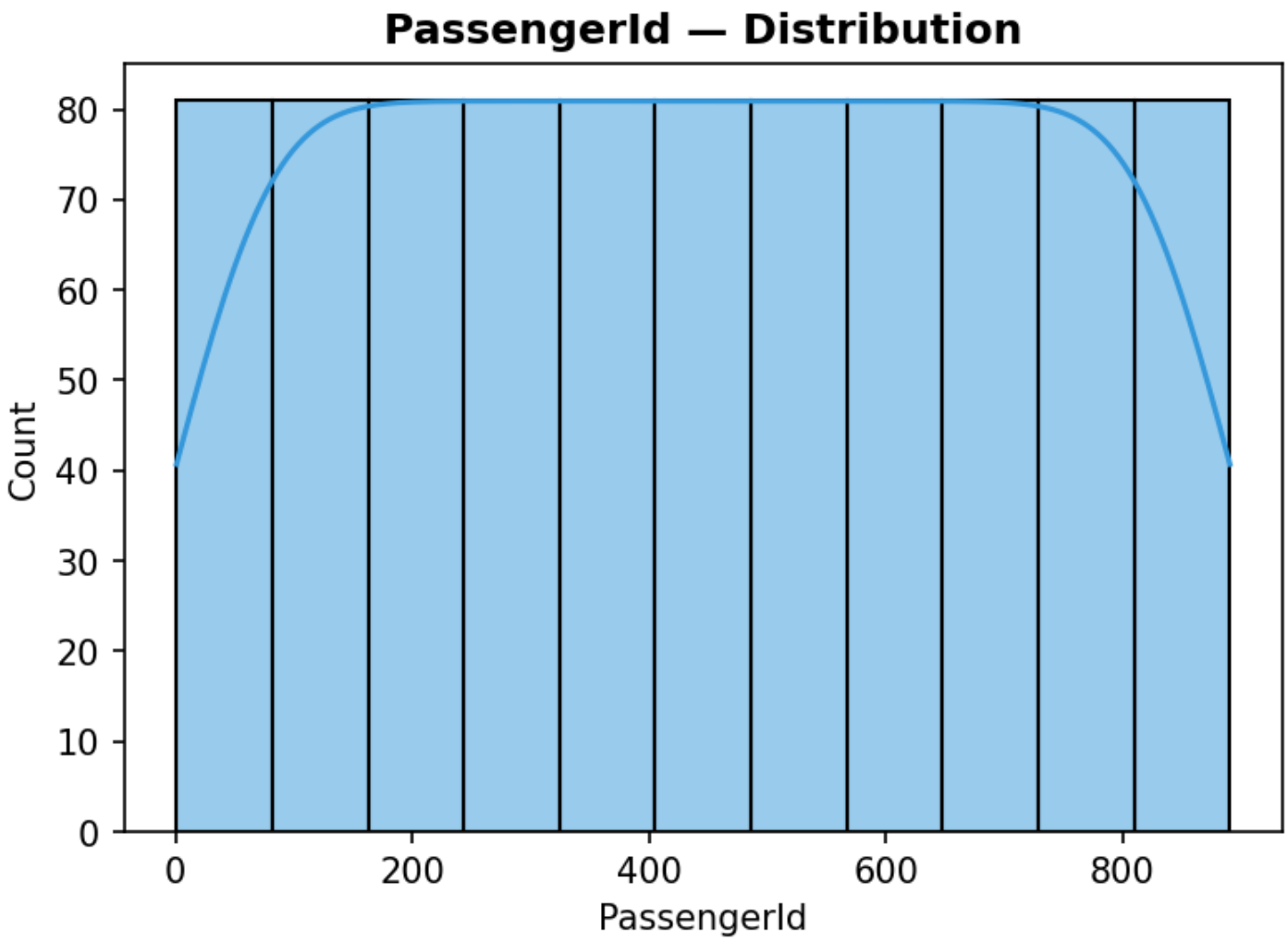
Column	Missing Count	Missing %
Age	177	19.87%
Cabin	687	77.1%
Embarked	2	0.22%

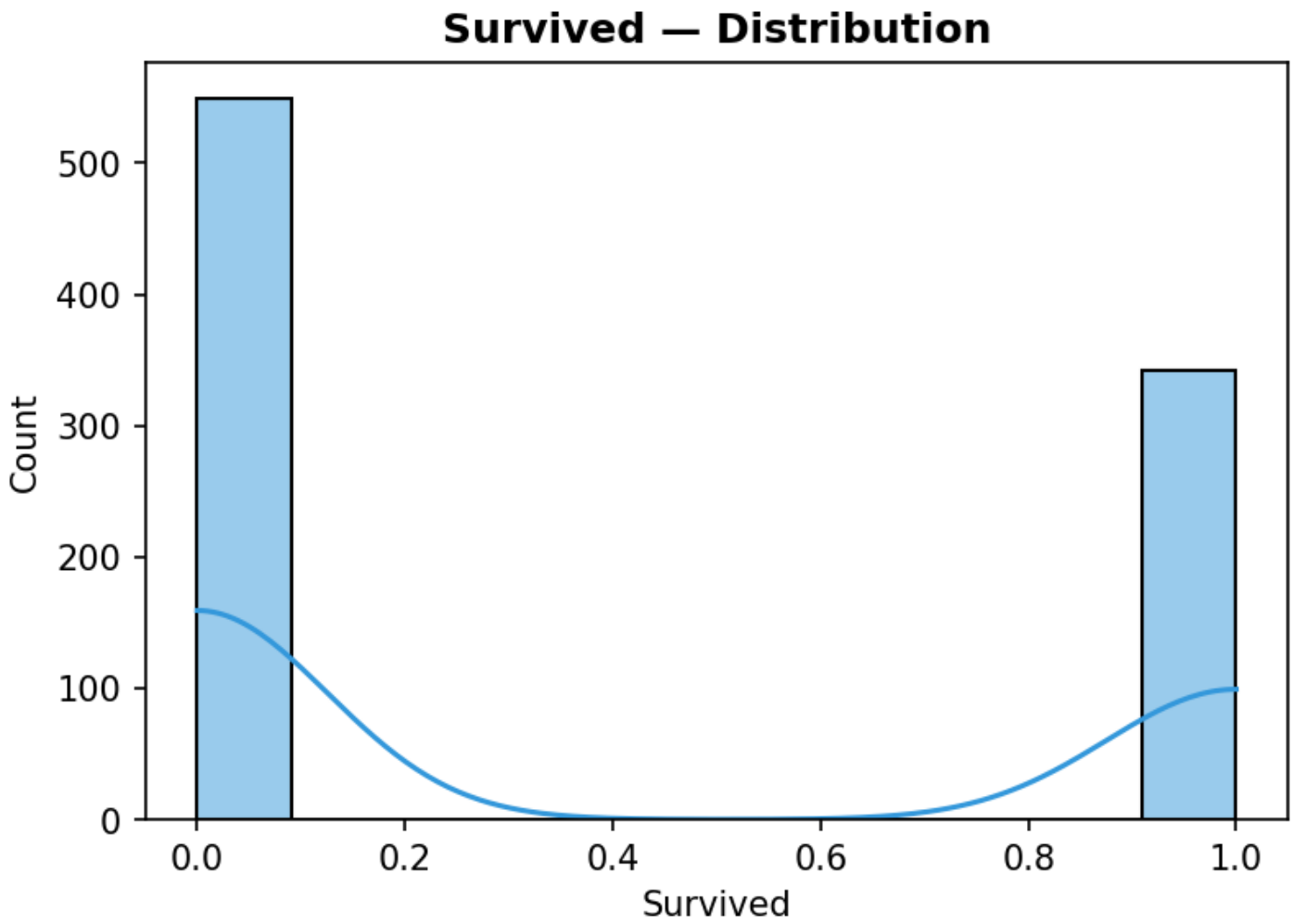


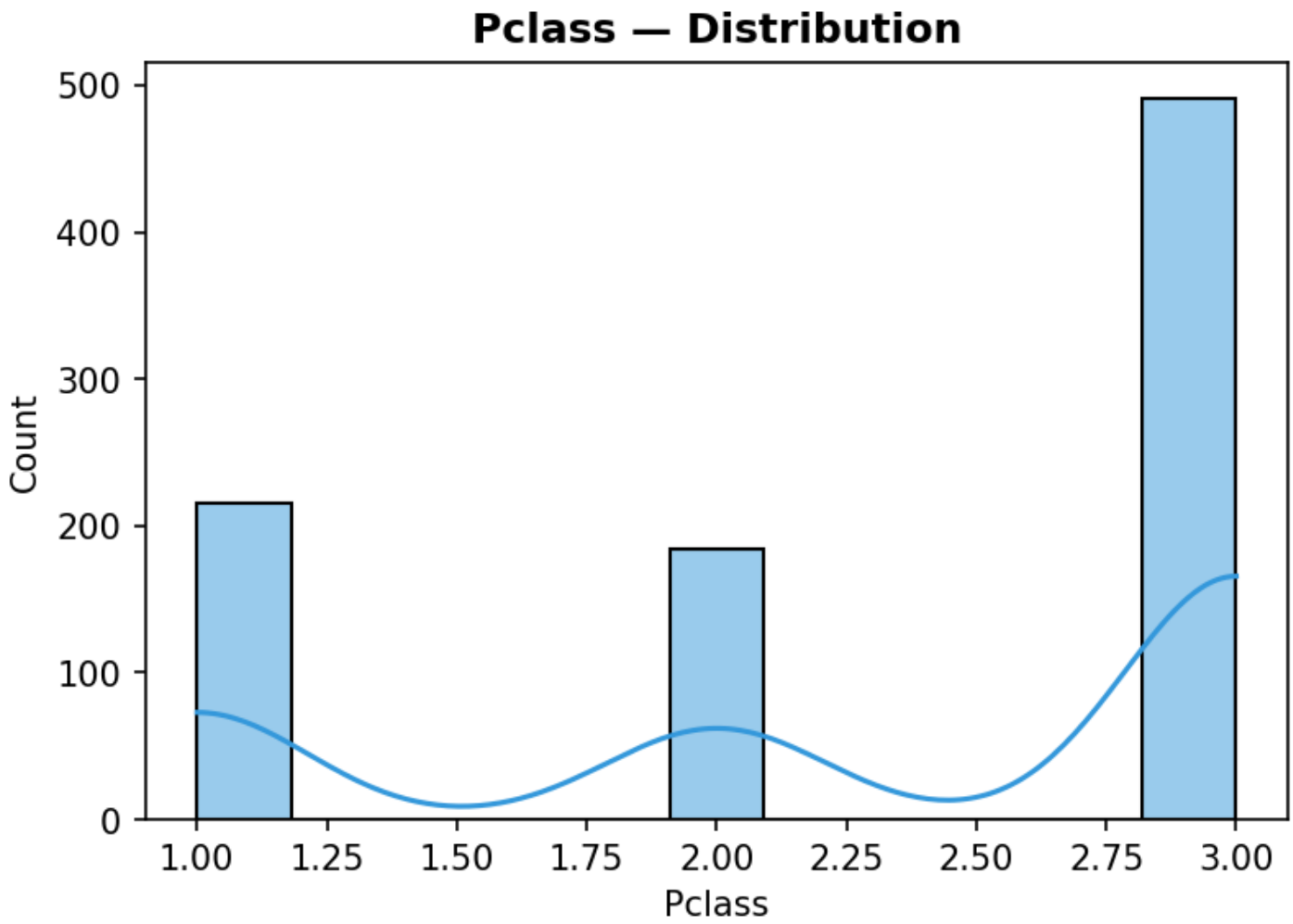

Missing Data by Column
50% line
Embarked
Cabin
Age
0
10
20
30
40
50
60
70
80
Missing %

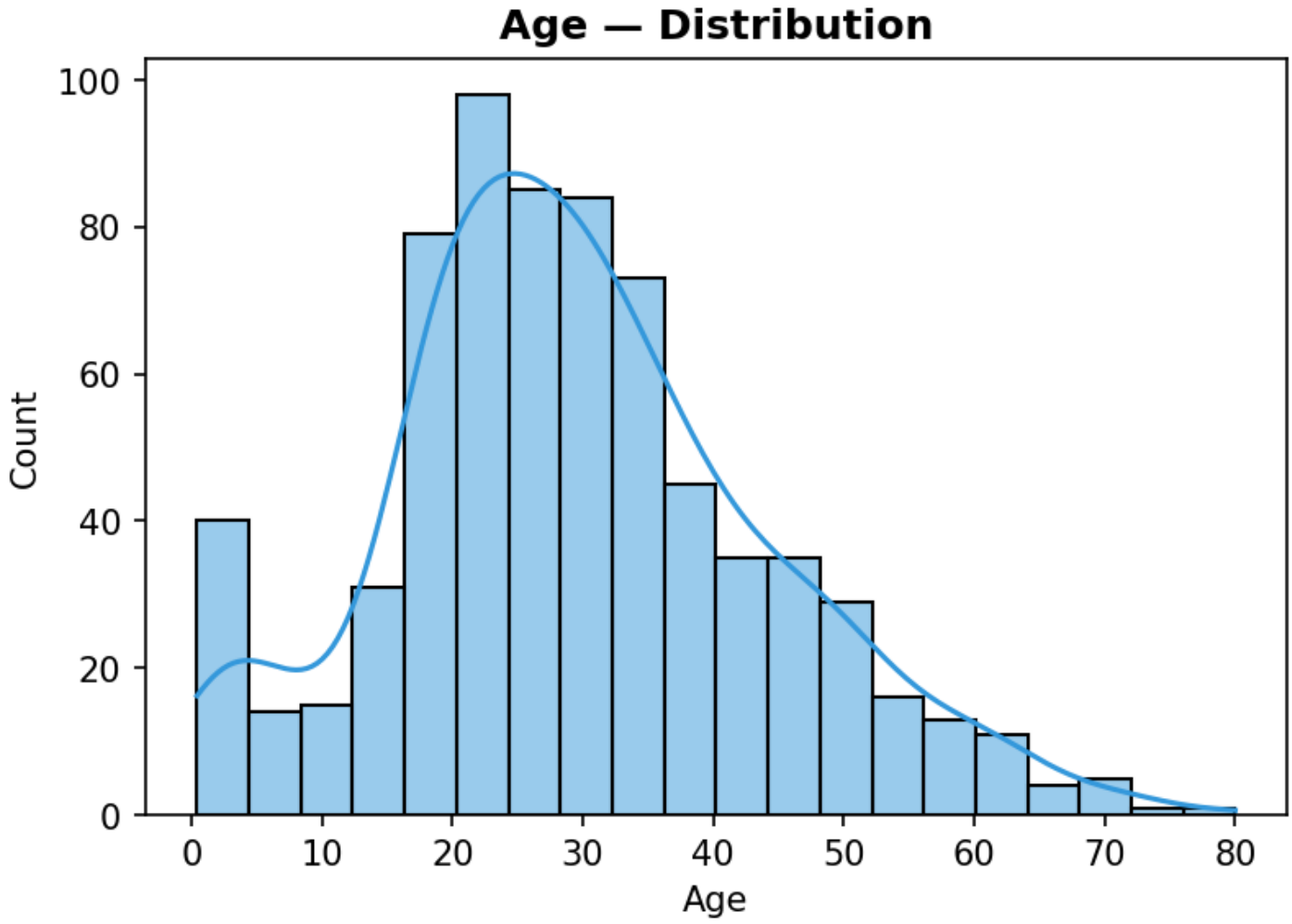
3. Distributions

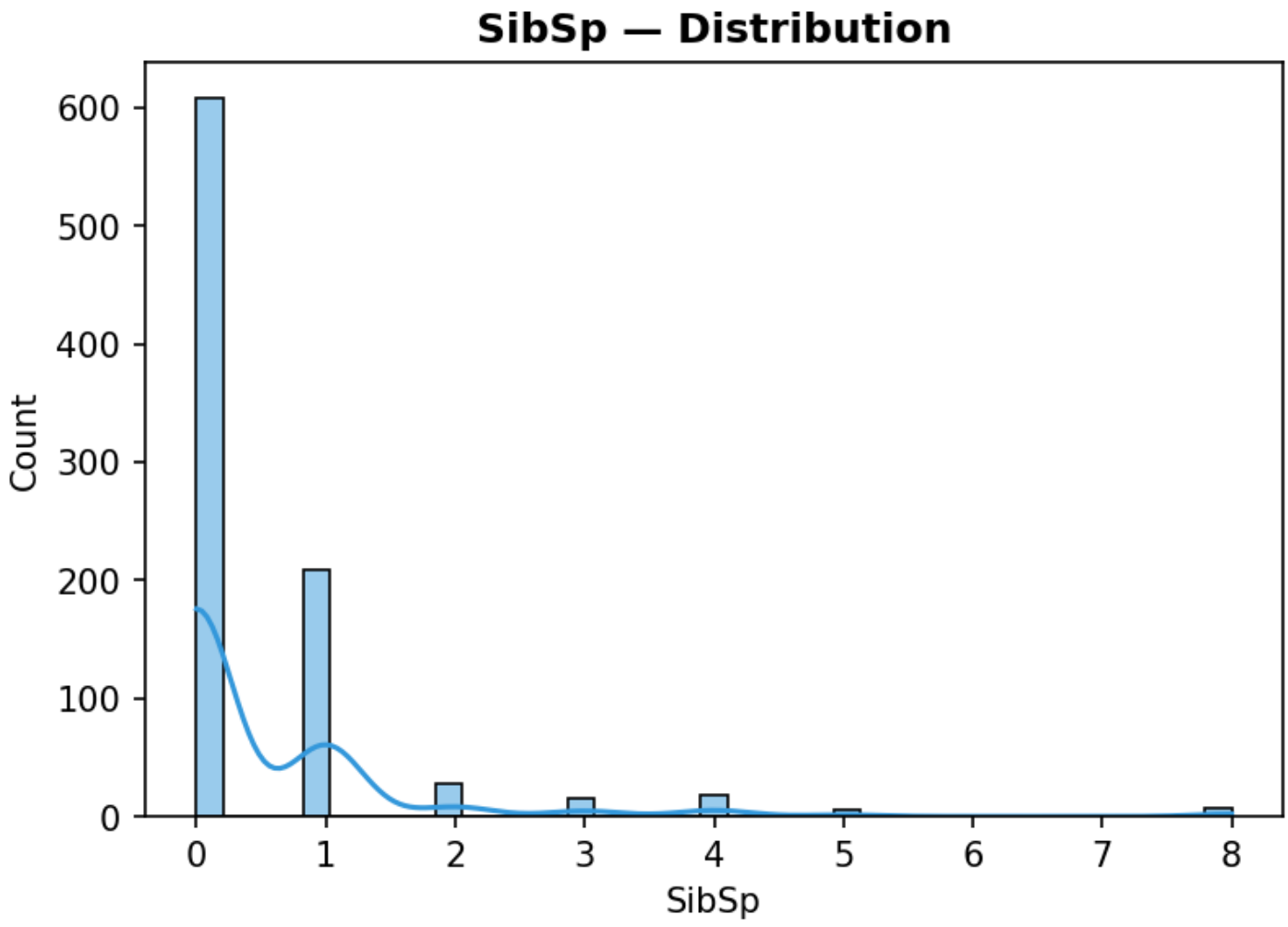
Numeric Columns

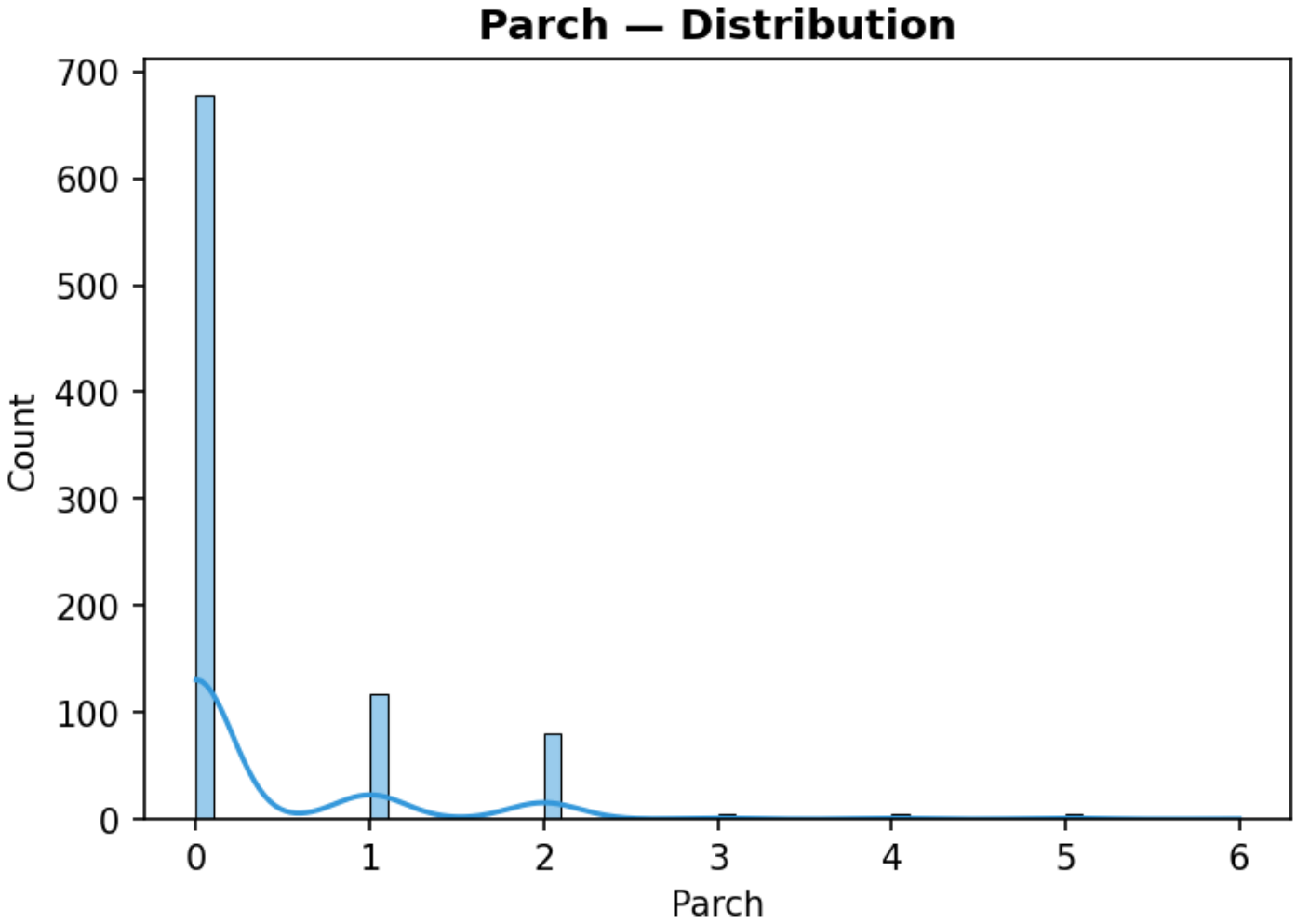


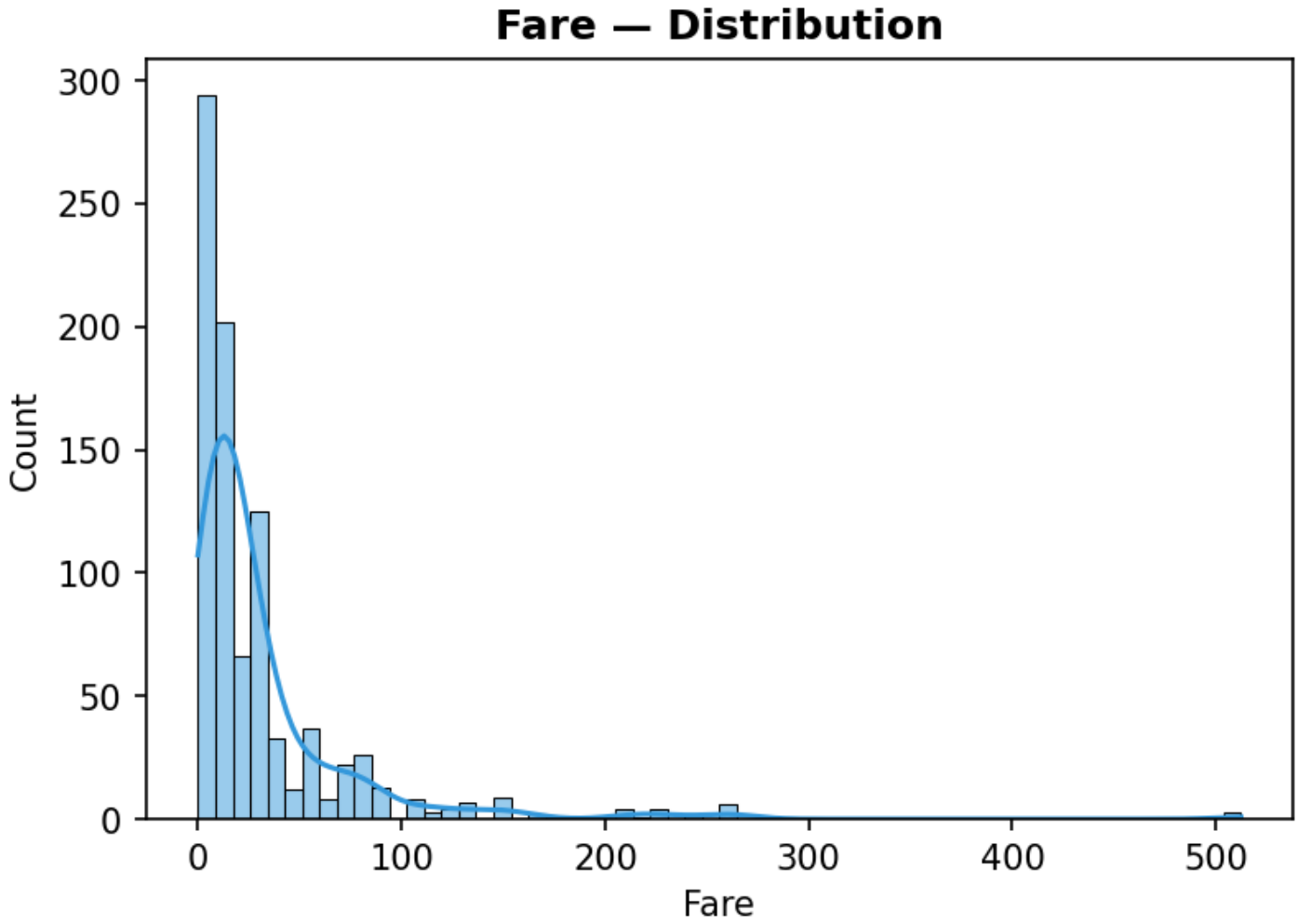

Survived — Distribution
Count
500
400
300
200
100
0
0.0
0.2
0.4
0.6
0.8
1.0
Survived


Pclass — Distribution
Count
500
400
300
200
100
0
1.00
1.25
1.50
1.75
2.00
2.25
2.50
2.75
3.00
Pclass

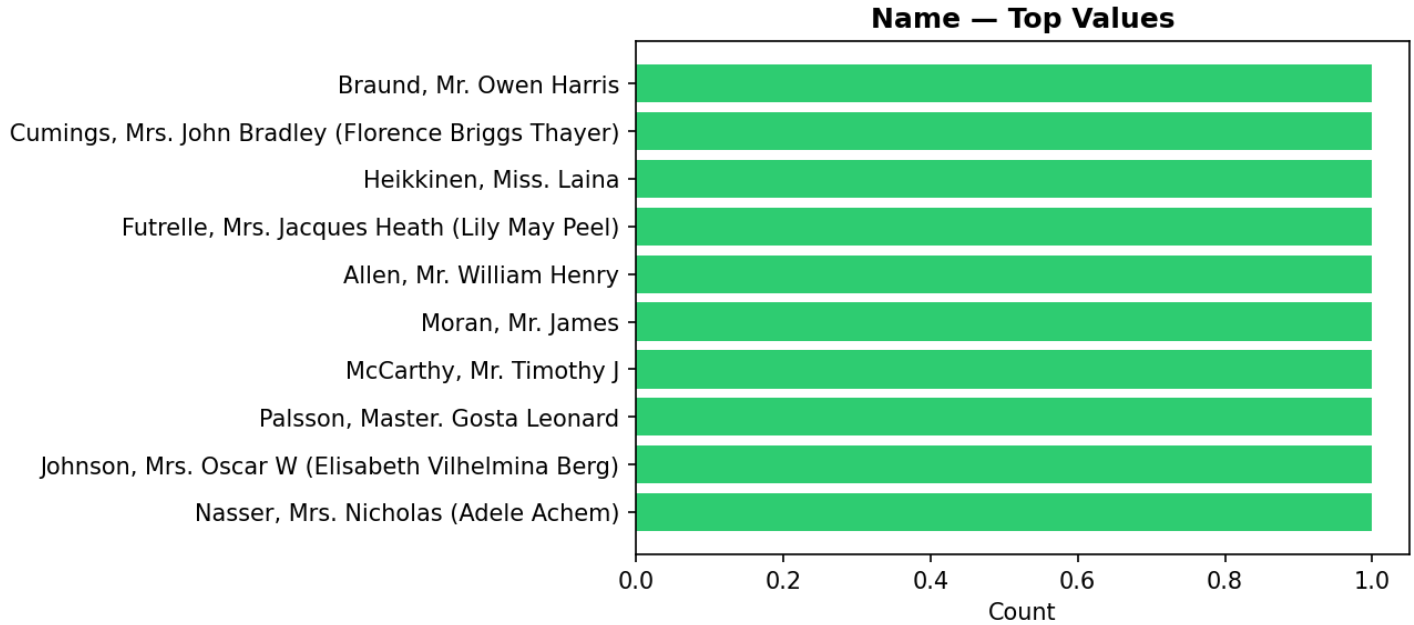

Age — Distribution
Count
Age
0
20
40
60
80
100
0
10
20
30
40
50
60
70
80

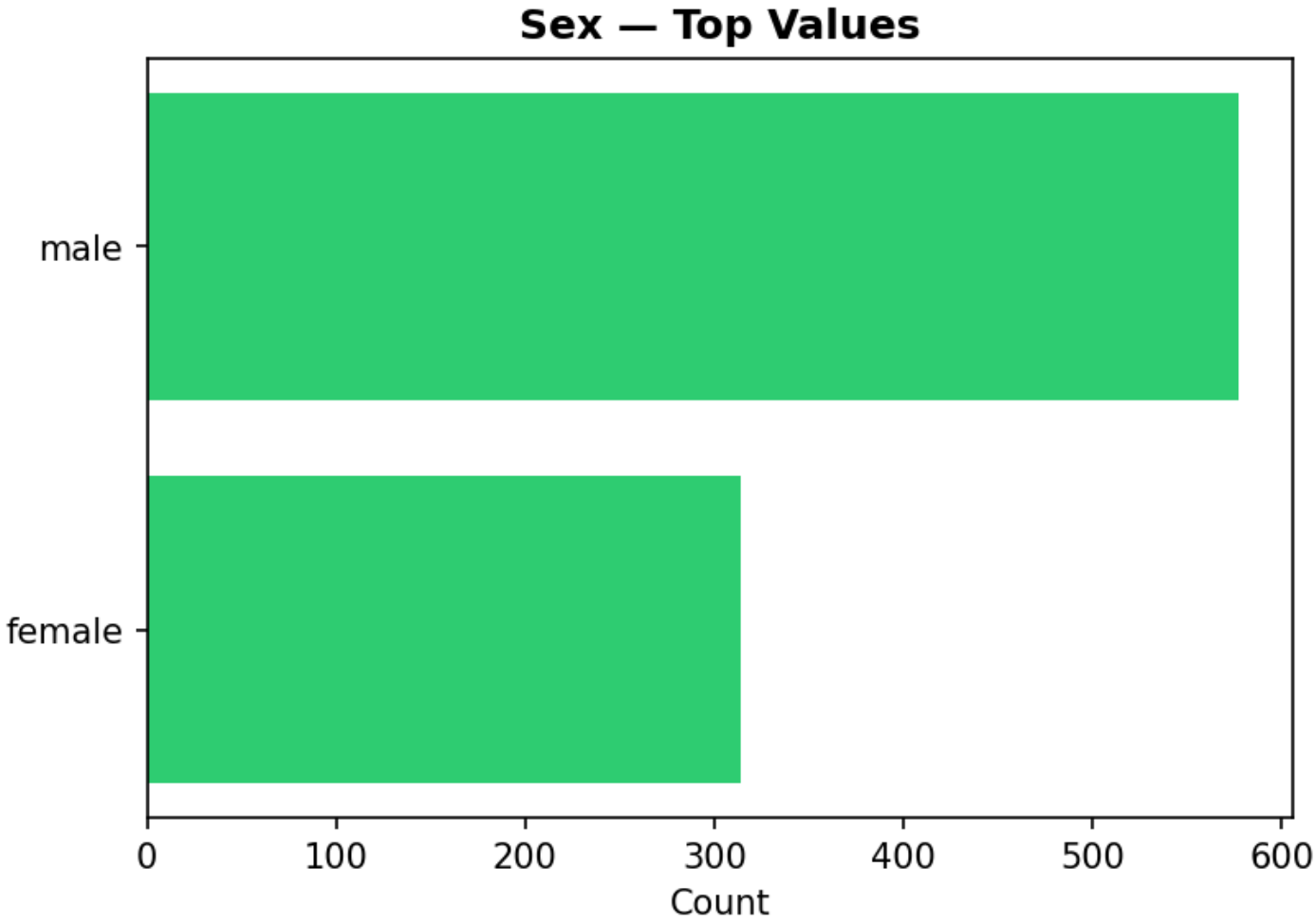

SibSp — Distribution
Count
600
500
400
300
200
100
0
0
1
2
3
4
5
6
7
8
SibSp

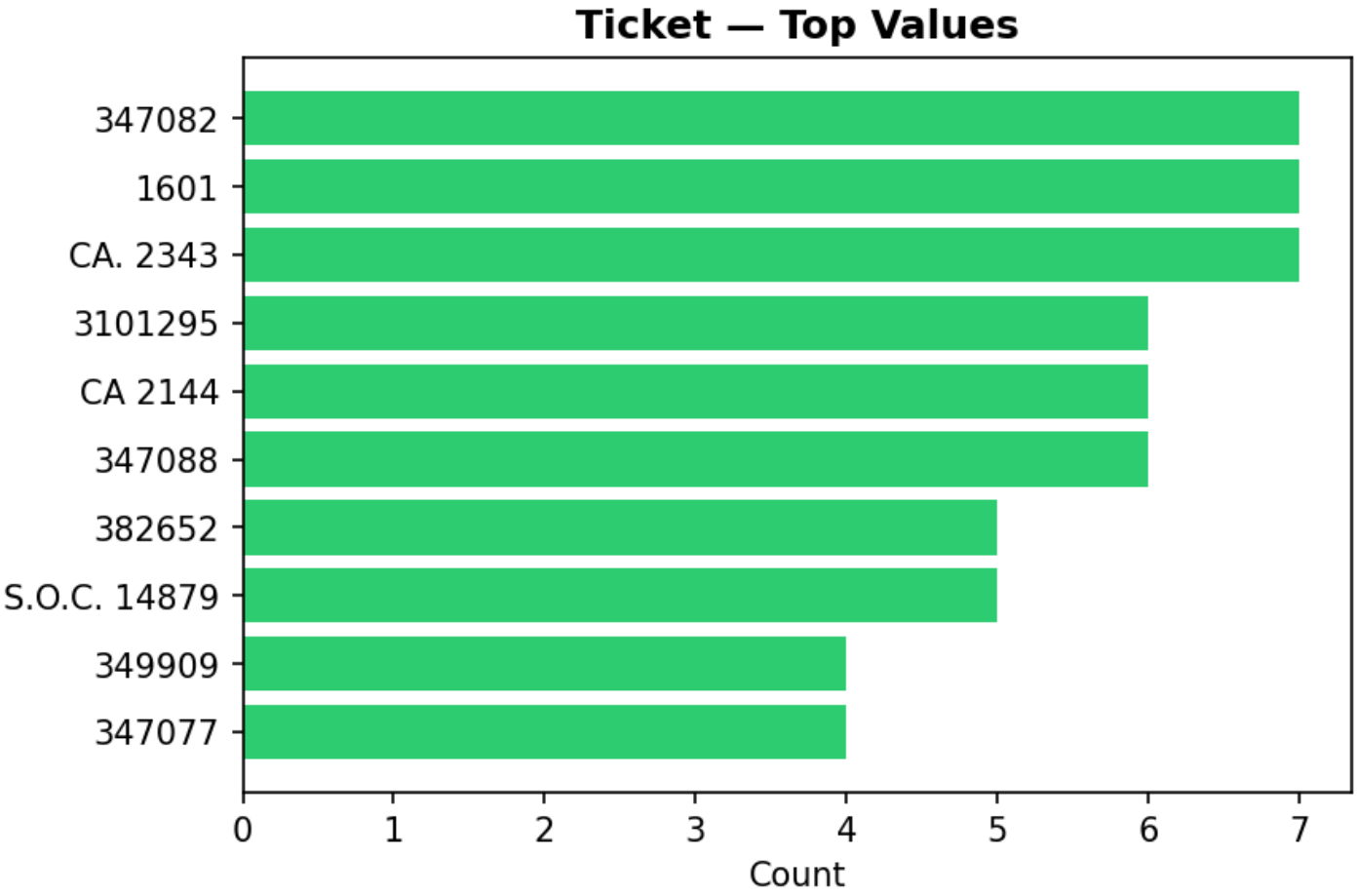

Parch — Distribution
Count
Parch
0
100
200
300
400
500
600
700
0
1
2
3
4
5
6

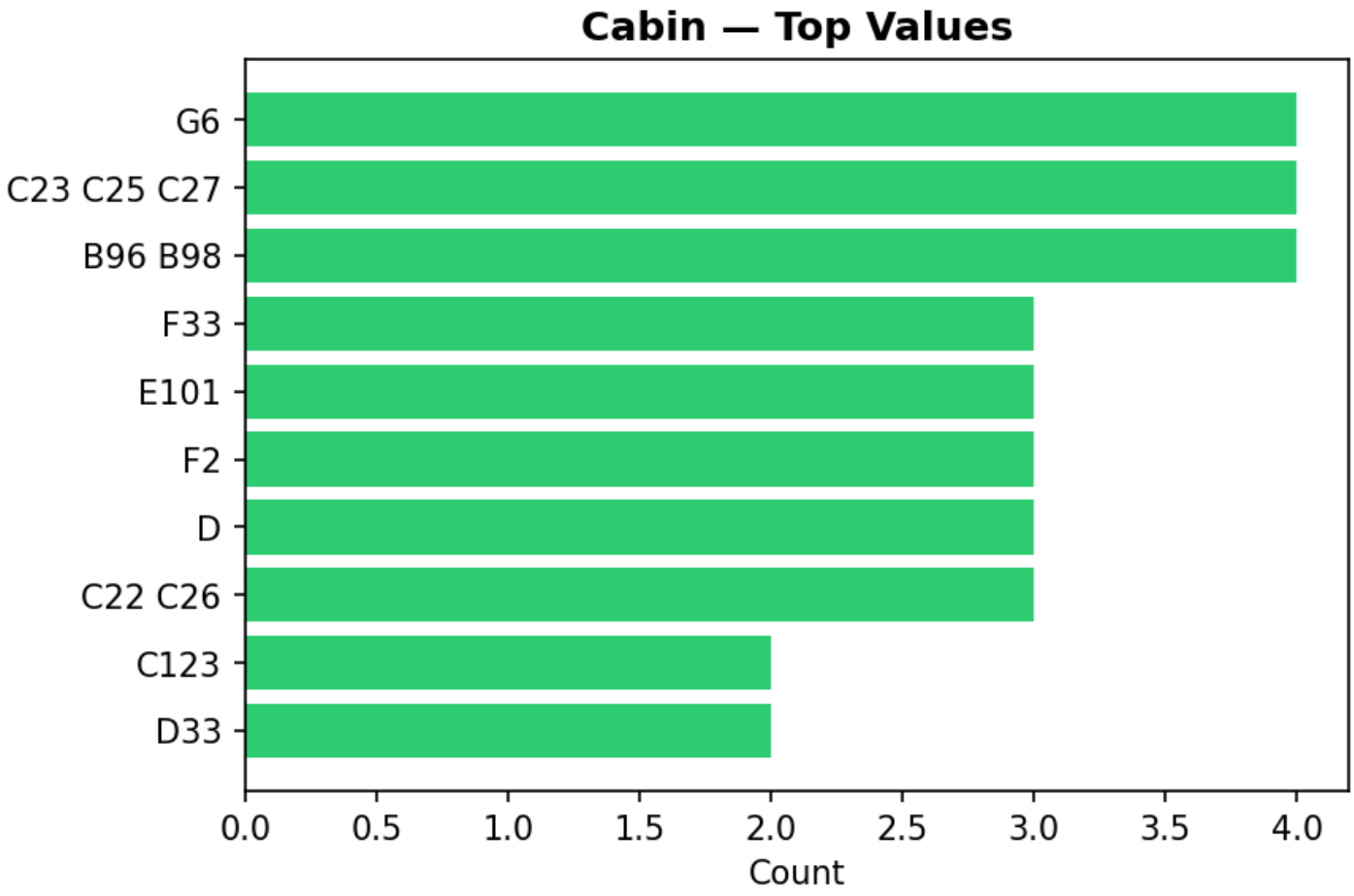


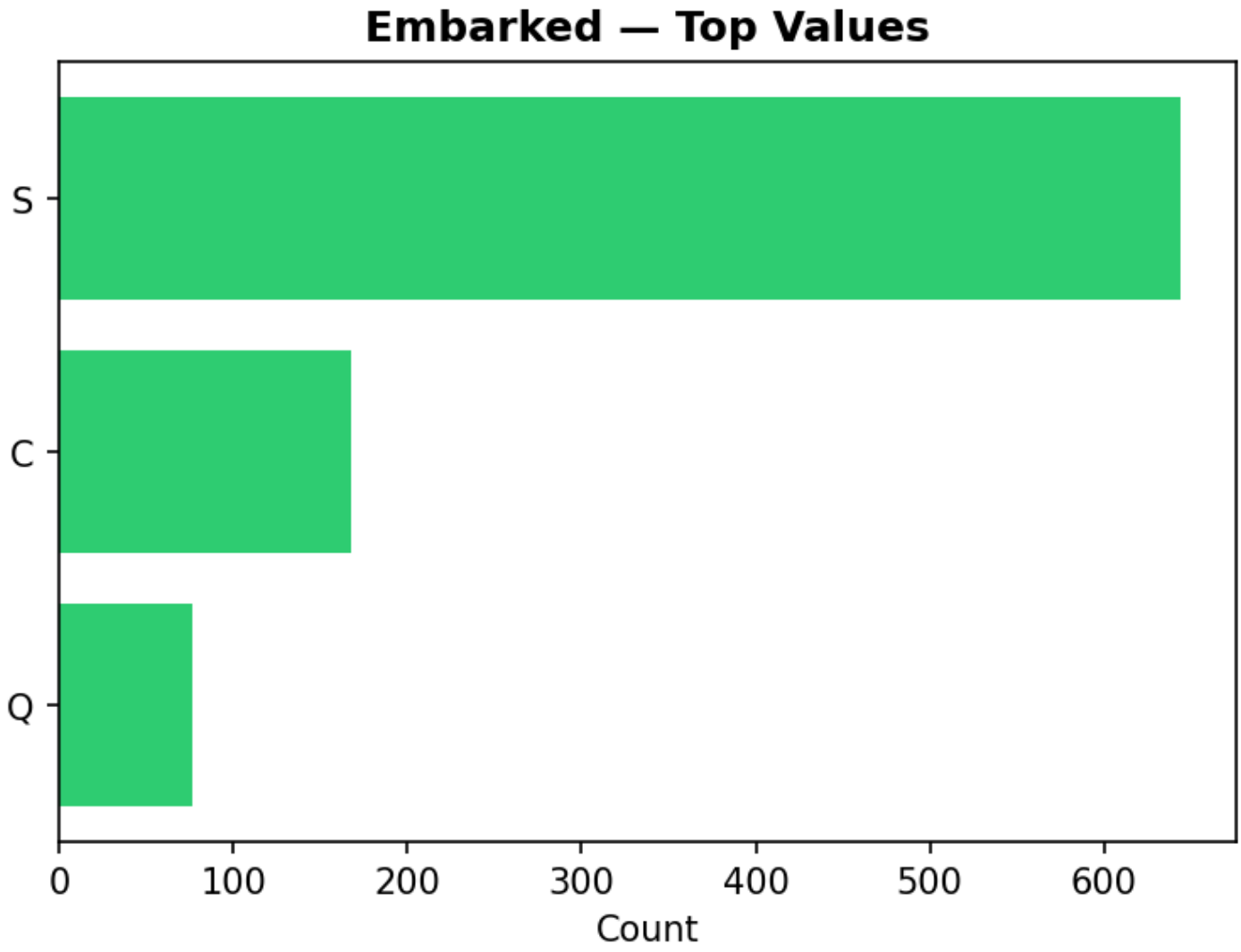
Categorical Columns




Sex — Top Values
male
female
0
100
200
300
400
500
600
Count

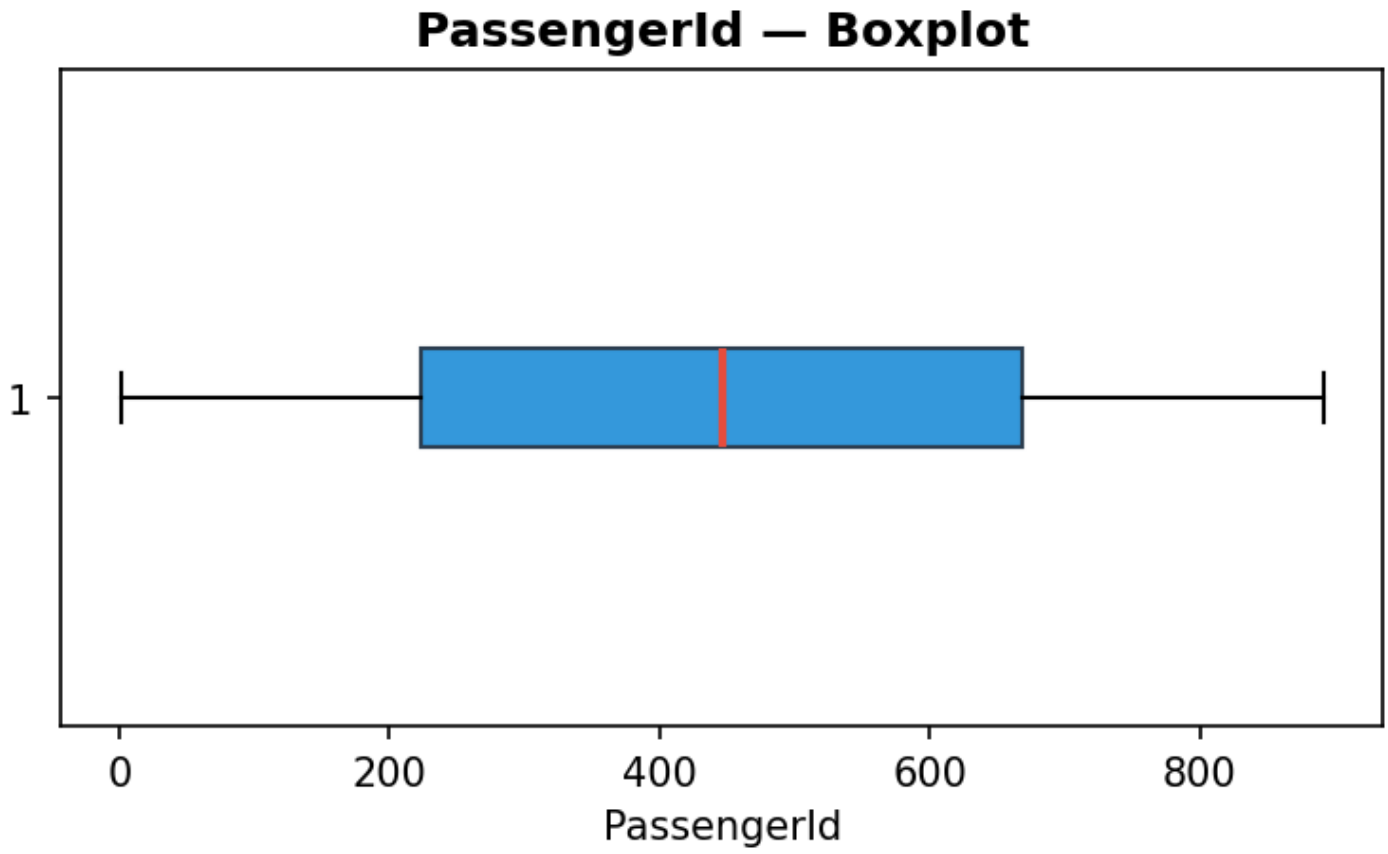

Ticket — Top Values
347082
1601
CA. 2343
3101295
CA 2144
347088
382652
S.O.C. 14879
349909
347077
0
1
2
3
4
5
6
7
Count

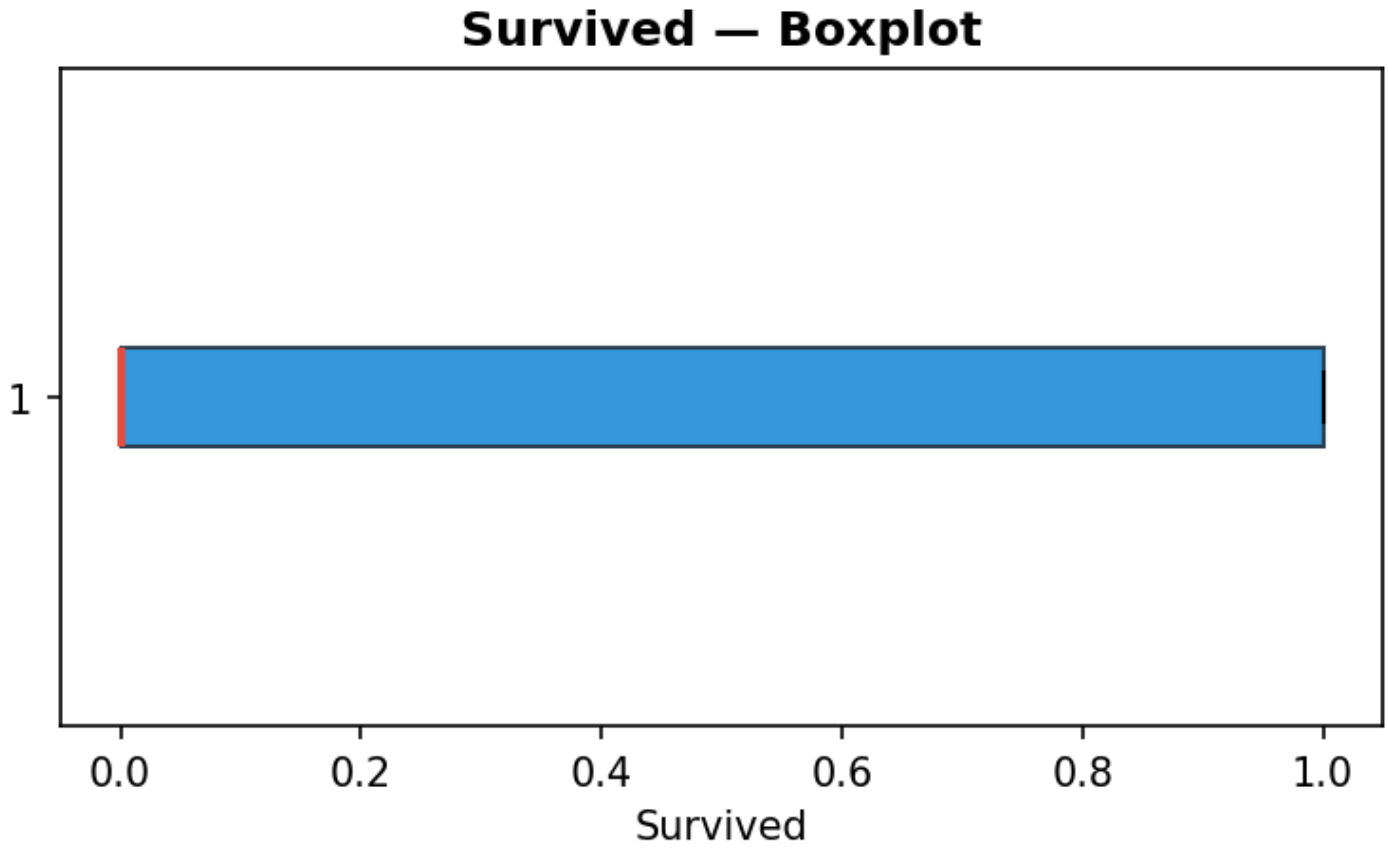

Cabin — Top Values
G6
C23 C25 C27
B96 B98
F33
E101
F2
D
C22 C26
C123
D33
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Count

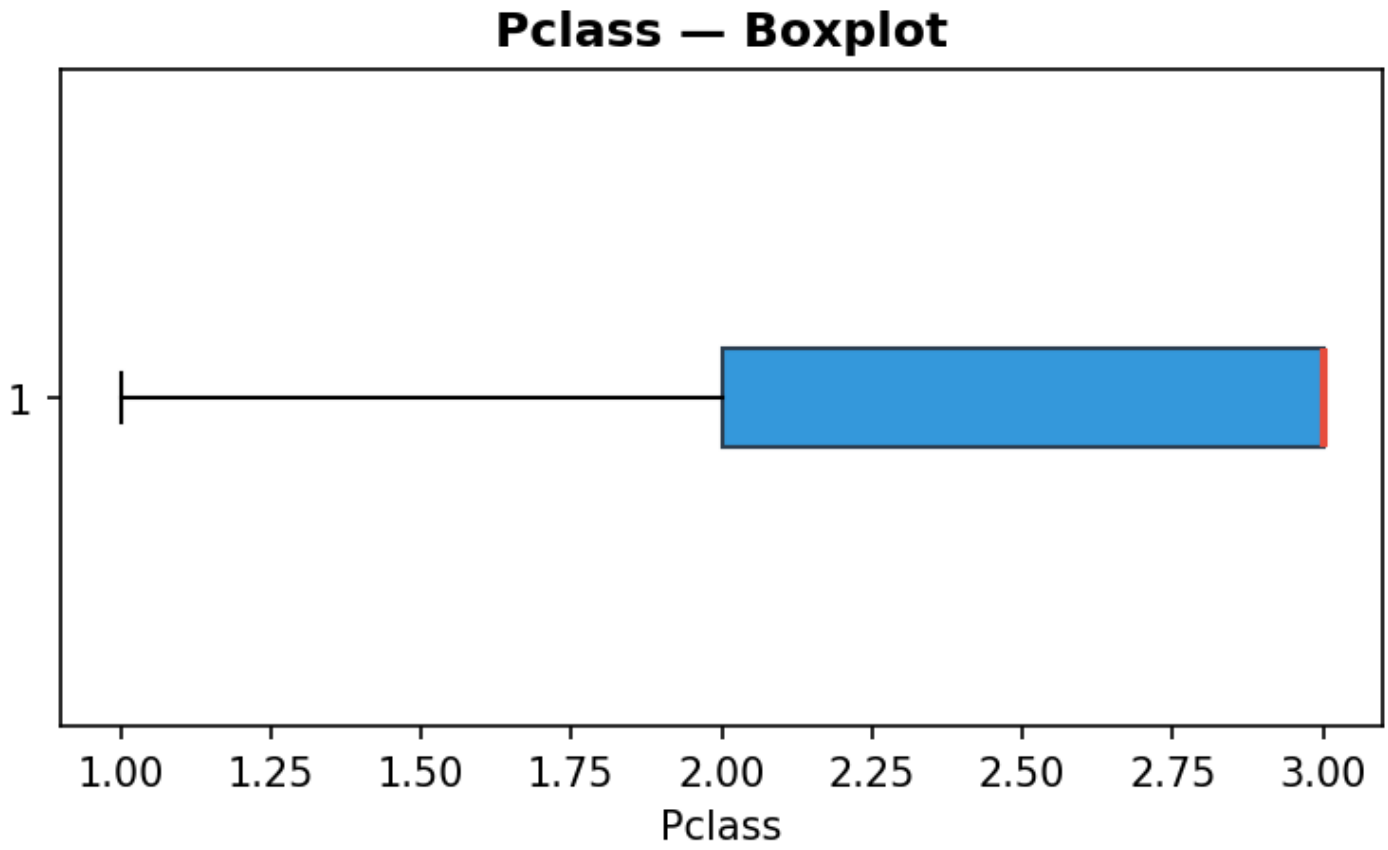

Embarked — Top Values
S
C
Q
0
100
200
300
400
500
600
Count

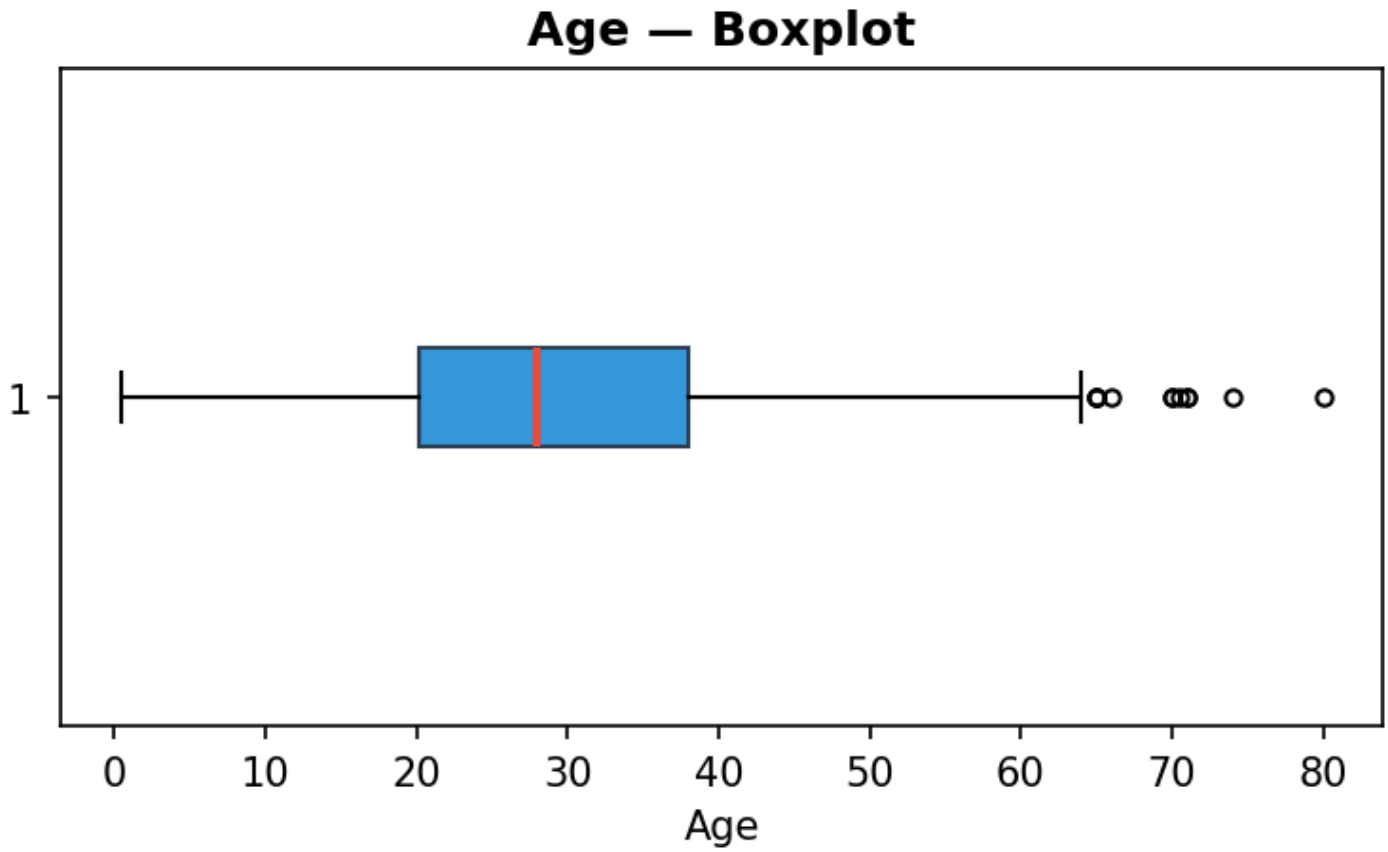
4. Outlier Detection

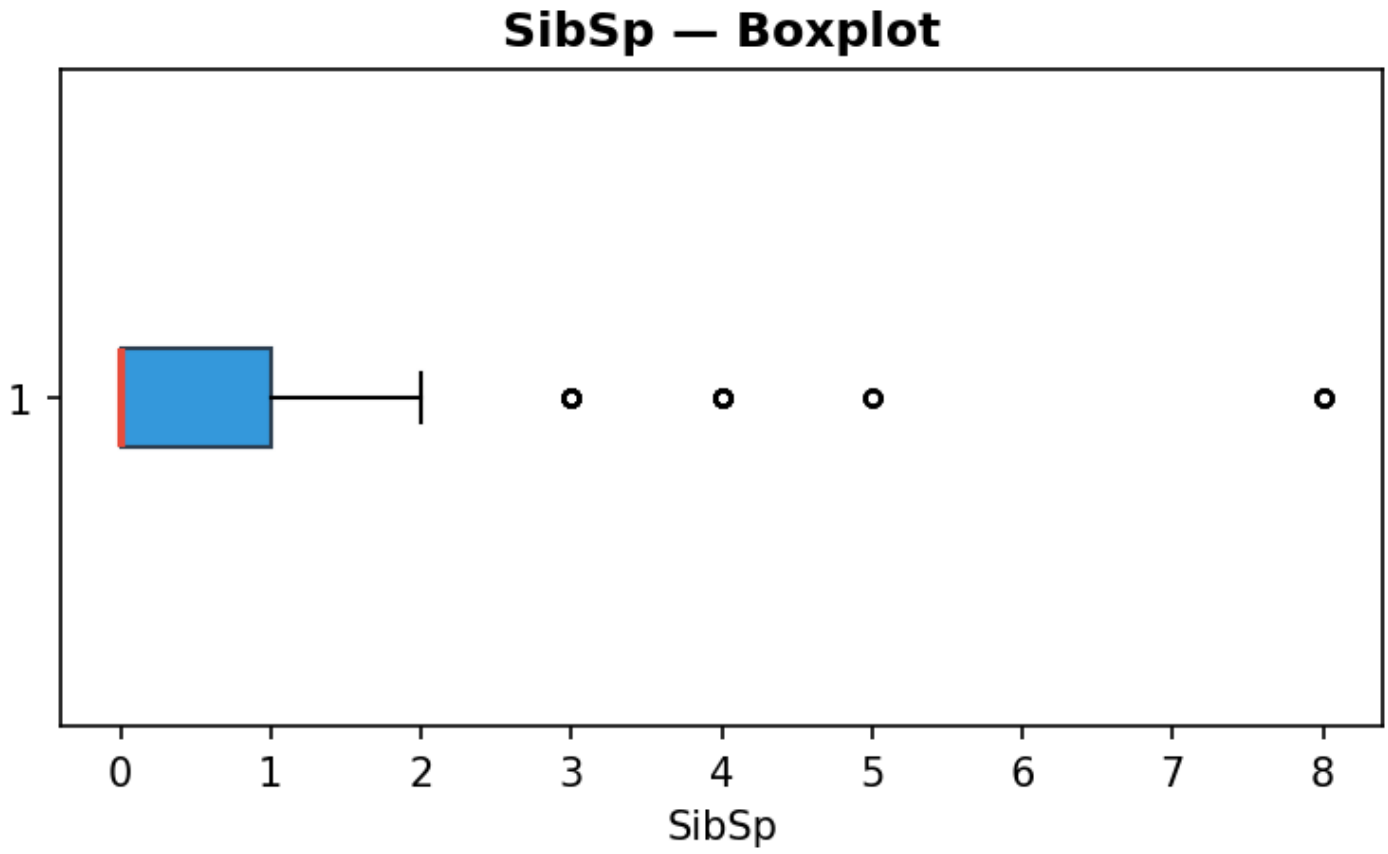
Column	Q1	Q3	IQR	Lower	Upper	Outliers	%
PassengerId	223.5	668.5	445.0	-444.0	1336.0	0	0%
Survived	0.0	1.0	1.0	-1.5	2.5	0	0%
Pclass	2.0	3.0	1.0	0.5	4.5	0	0%
Age	20.125	38.0	17.875	-6.6875	64.8125	11	1.54%
SibSp	0.0	1.0	1.0	-1.5	2.5	46	5.16%
Parch	0.0	0.0	0.0	0.0	0.0	213	23.91%
Fare	7.9104	31.0	23.0896	-26.724	65.6344	116	13.02%

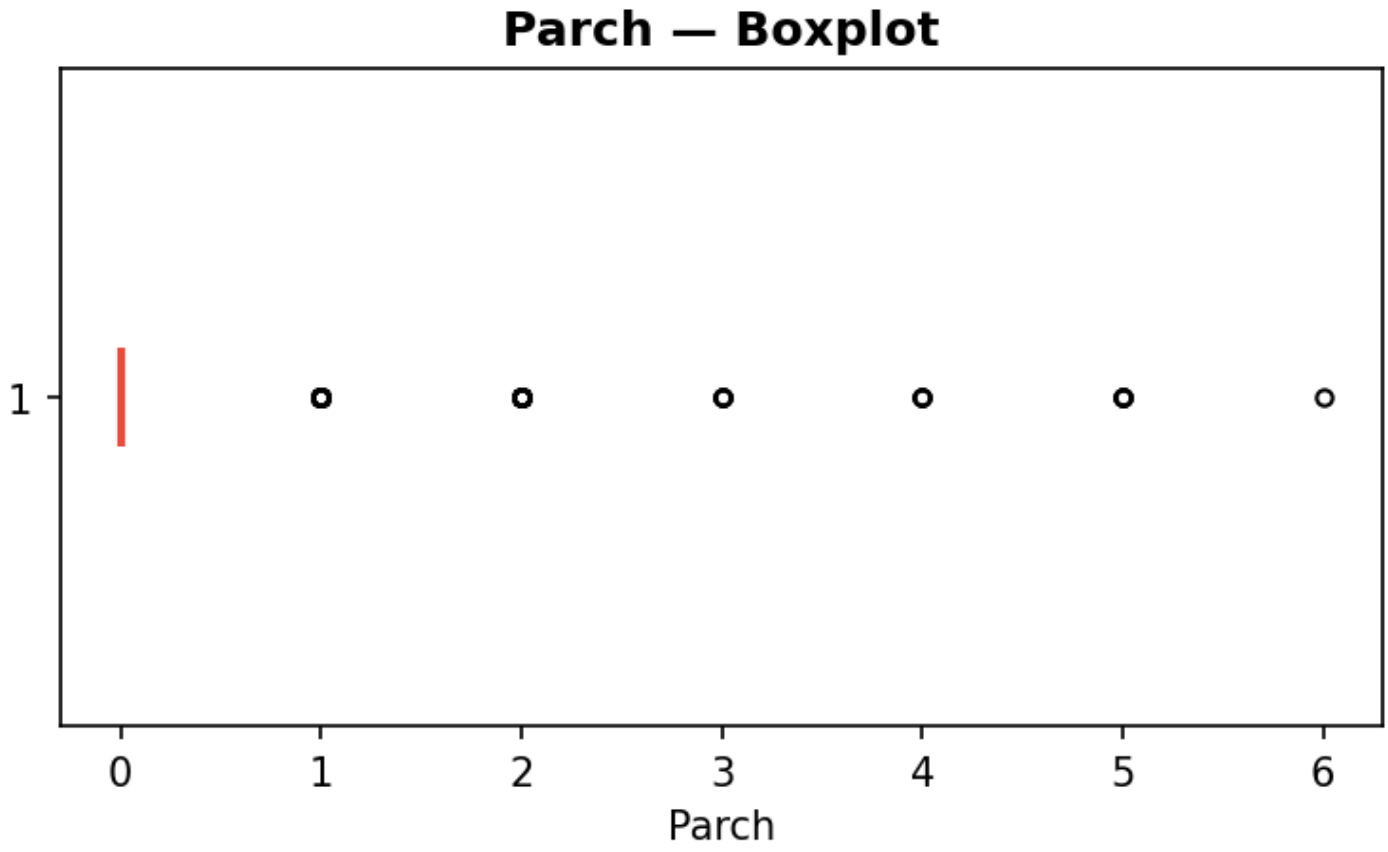

PassengerId — Boxplot
1
0
200
400
600
800
PassengerId

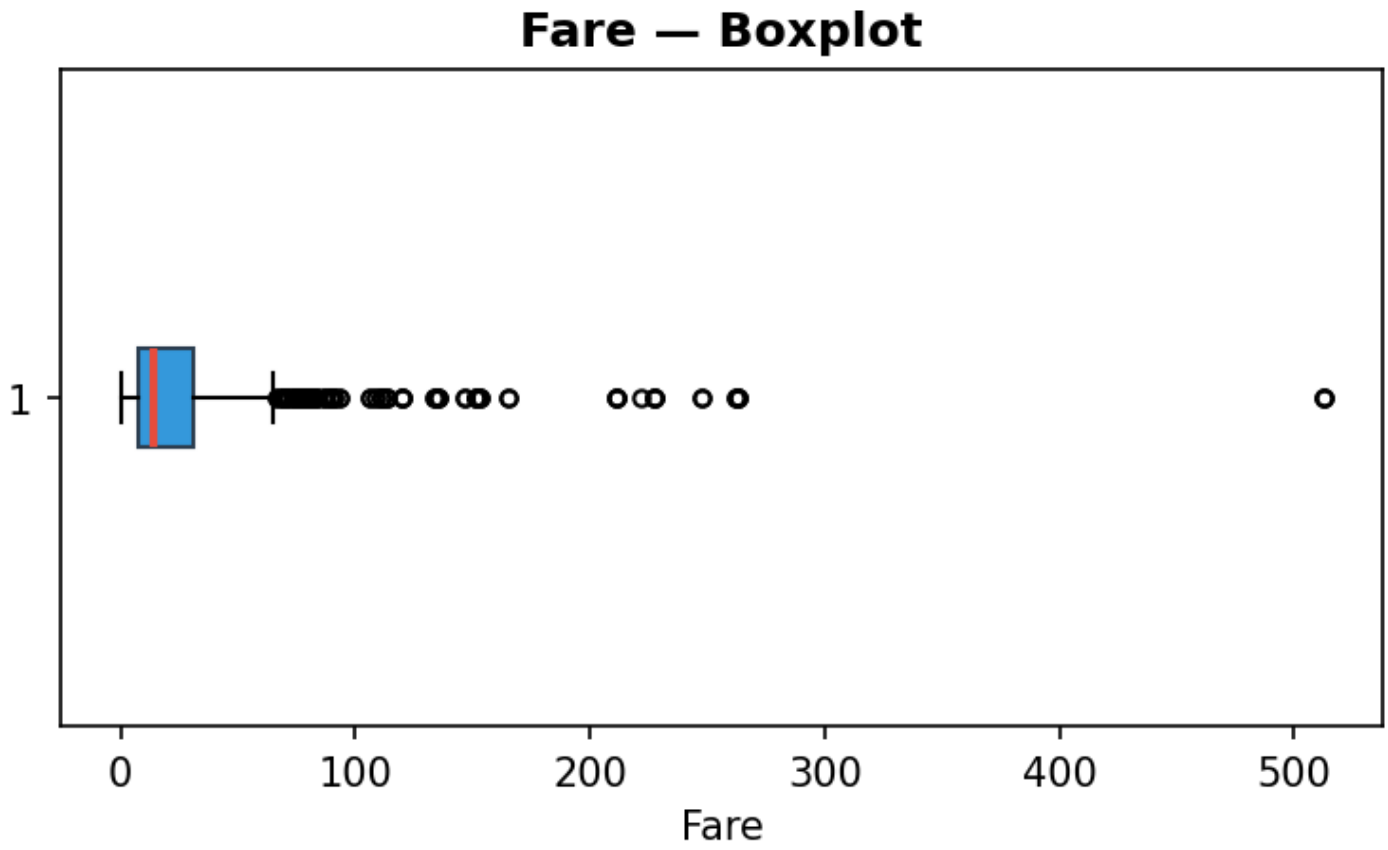

Survived — Boxplot
1
0.0
0.2
0.4
0.6
0.8
1.0
Survived


Pclass — Boxplot
1
1.00
1.25
1.50
1.75
2.00
2.25
2.50
2.75
3.00
Pclass

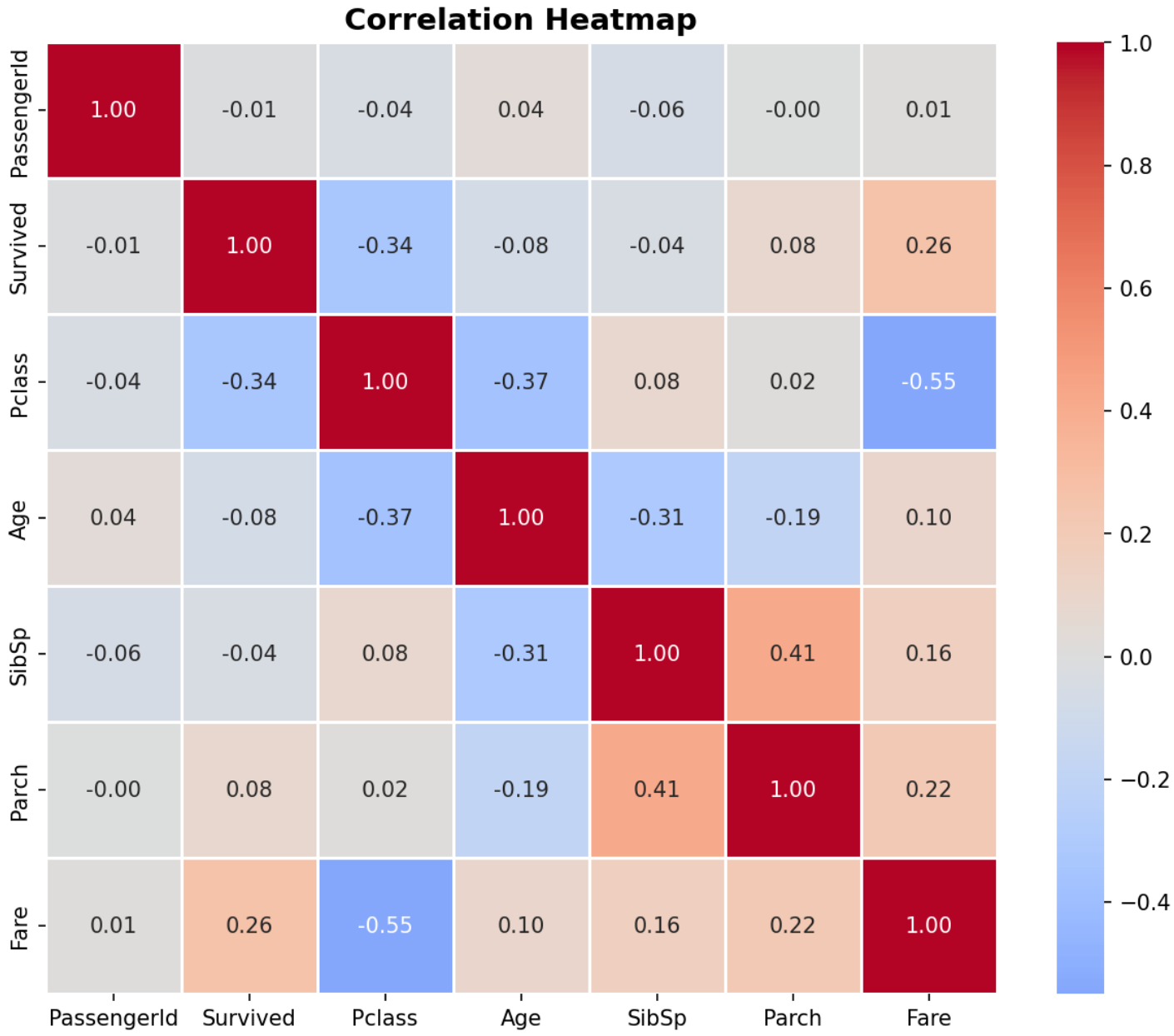

Age — Boxplot
1
0
10
20
30
40
50
60
70
80
Age


SibSp — Boxplot
1
0
1
2
3
4
5
6
7
8
SibSp


Parch — Boxplot
1
0
1
2
3
4
5
6
Parch


Fare — Boxplot
1
0
100
200
300
400
500
Fare

5. Correlation Analysis



Top Correlated Pairs

Column A	Column B	Correlation
Pclass	Fare	-0.5495
SibSp	Parch	0.4148
Pclass	Age	-0.3692
Survived	Pclass	-0.3385

Age	SibSp	-0.3082
Survived	Fare	0.2573
Parch	Fare	0.2162
Age	Parch	-0.1891
SibSp	Fare	0.1597
Age	Fare	0.0961

6. Statistical Summary

Column	Count	Mean	Median	Std	Min	Max	Skewness	Kurtosis
PassengerId	891	446.0	446.0	257.3538	1.0	891.0	0.0	-1.2
Survived	891	0.3838	0.0	0.4866	0.0	1.0	0.4777	-1.7718
Pclass	891	2.3086	3.0	0.8361	1.0	3.0	-0.6295	-1.2796
Age	714	29.6991	28.0	14.5265	0.42	80.0	0.3883	0.1686
SibSp	891	0.523	0.0	1.1027	0.0	8.0	3.6891	17.7735
Parch	891	0.3816	0.0	0.8061	0.0	6.0	2.7445	9.7166
Fare	891	32.2042	14.4542	49.6934	0.0	512.3292	4.7793	33.2043